Project:	VT CH3 SVC	Topic: Functional Safety Requirement for HOST SYSTEM	Date: 2025-10-22	Revision:	17.2

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_227	**1**	**Scope**	**Heading**	**n/a**	**n/a**
ECU_RS_269	1.0-1	Purpose of this document is to describe the safety requirements allocated to the HOST SYSTEM and requirements to external interfaces for the project VT CH3 SVC.	Description	n/a	n/a
ECU_RS_373	1.0-2	The content of this document is Bosch proprietary and it is confidential. May not be made accessible for an unauthorized third party.	Description	n/a	n/a
ECU_RS_376	1.0-3	Functional Safety Manager: Amanda Leila AOUASSAR (XC-AC/ECT2) Functional Safety Hardware Expert: Racz Tibor (XC-AC/ENH2-Bp)	Description	n/a	n/a
ECU_RS_377	**1.1**	**Abbreviations**	**Heading**	**n/a**	**n/a**
ECU_RS_378	1.1.0-1	EEPROM = Electrically Erasable Programmable Read-Only Memory FuSa = Functional Safety GMSL = Gigabit Multimedia Serial Link (Analog Devices) HFPN = Horizontanl Fixed Pattern Noise (related to Imager chip) ISP = Image Signal Processing PMIC = Power Management IC PoC = Power Over Coax SoC = System On Chip n/a = not applicable	Description	n/a	n/a
ECU_RS_379	1.1.0-2	AoU = Assumption of Use E/E = Electrical or Electronic FHTI = Fault Handling Time Interval FRTI = Fault Reaction Time Interval FTTI = Fault Tolerant Time Interval IF = Intended Functionality SM = Safety Mechanism TSC = Technical Safety Concept TSR = Technical Safety Requirement	Description	n/a	n/a
ECU_RS_450	**1.2**	**Definitions**	**Heading**	**n/a**	**n/a**
ECU_RS_422	1.2.0-1	OVERALL SYSTEM = The union of the HOST SYSTEM and CAMERA HEAD that implements the vehicle level function. HOST SYSTEM = The system(s) that control the CAMERA HEAD and receive its video stream. It can be a single ECU or multiple ECUs. CH = CAMERA HEAD	Description	n/a	n/a
ECU_RS_514	1.2.0-2	OFF time = the Camera HEAD is not powered. ON time = the Camera HEAD is electrically powered. not defined whether it streams video or not.	Description	n/a	n/a
ECU_RS_515	**1.2.1**	**Attribute Definitions**	**Heading**	**n/a**	**n/a**
ECU_RS_516	**1.2.1.1**	**Object_Type**	**Heading**	**n/a**	**n/a**
ECU_RS_517	1.2.1.1.0-1	Possible values: • "Heading": the headers in the document structure. Not a requuirement, only for documentation purposes. Implementation/verification is not applicable. • "Description": Additional information that help to understand the items. context. meaning/operation of the function. Implementation/verification is not applicable. • "Requirement": Requirement to be implmemented by the HOST SYSTEM.	Description	n/a	n/a
ECU_RS_283	**1.3**	**System Identification**	**Heading**	**n/a**	**n/a**
ECU_RS_284	1.3.0-1	Customer: Volvo Trucks Project: VT CH3 SVC Application: Surround View Camera System Area of usage: commercial vehicles SOP: 2027	Description	n/a	n/a
ECU_RS_231	**1.4**	**System Overview and Boundaires**	**Heading**	**n/a**	**n/a**
ECU_RS_467	1.4.0-1	Intended usage: create image about the scene and send to HOST SYSTEM via high-speed serial link.	Description	n/a	n/a
ECU_RS_466	1.4.0-2	CAMERA HEAD consists of an image sensor. serializer. PMIC and an EEPROM. It is connected to the HOST SYSTEM via a Co-axial cable with a coaxial connector.	Description	n/a	n/a
ECU_RS_468	1.4.0-3	* Image data from camera head to HOST SYSTEM via the co-axial cable. * Bidirectional control data (I2C messages) are routed via the Co-axial cable.	Description	n/a	n/a
ECU_RS_490	1.4.0-4	Camera Head - Architectural Design of Electronics	Description	n/a	n/a

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_445	1.4.0-5	The Camera Head is designed to FAIL SAFE reaction in case of any safety relevant failures.	Description	n/a	n/a
ECU_RS_374	**1.5**	**Responsibilites**	**Heading**	**n/a**	**n/a**
ECU_RS_446	1.5.0-1	Bosch is responsible for: - PMIC Configuration Settings - Imager Configuration Settings - EEPROM content Volvo is responsible for: - Serializer configuration	Description	n/a	n/a
ECU_RS_228	**2**	**Technical Requirements**	**Heading**	**n/a**	**n/a**
ECU_RS_233	**2.1**	**Technical Safety Requirements**	**Heading**	**n/a**	**n/a**
ECU_RS_327	2.1.0-1	This section contains detailed specifications of the requirements that are allocated for the HOST SYSTEM. These requirements shall be implemented by the HOST SYSTEM. Further refinement into HW and SW requirement is not given as it is in the responsibility of the developer of HOST SYSTEM and Bosch does not define constrains for implementation.	Description	n/a	n/a
ECU_RS_282	2.1.0-2	These include - measures required to achieve. maintain or restore the safety-related functions and properties of the item and its elements during production. service or decommissioning. - measures to internal and external production. Hint: measures to external production (at OEM plant) are described in the corresponding Technical Customer Documentation (TCD) and delivered to the OEM.	Description	n/a	n/a
ECU_RS_358	**2.1.1**	**Intrinsic Calibration Data**	**Heading**	**n/a**	**n/a**
ECU_RS_470	2.1.1.0-1	Intrinsic Calibration Data is used to correct certain distorsions of the optical system.	Description	n/a	n/a
ECU_RS_510	2.1.1.0-2	WHEN: once in each life cycle before the video stream is used for safety related purposes (start-up. restarts)	Requirement	IF	TBD
ECU_RS_345	2.1.1.0-3	The HOST SYSTEM shall read the Intrinsic Calibration Data from CAMERA HEAD. The Intrinsic Calibration Data stored in the EEPROM in the following memory sector(s): Document: [2] (Camera Interface Specification) Section: 5.3. EEPROM MAP Block Name: Intrinsic data block (F-Theta) If a contradiction recognized compared to other information source. Bosch shall be involved and clarify the point.	Requirement	IF	TBD
ECU_RS_353	2.1.1.0-4	The HOST SYSTEM shall read out and check CRC Checksum of the Intrinsic Calibration Data before using it. The Checksum is stored in the EEPROM together with Calibration data.	Requirement	SM	TBD
ECU_RS_333	2.1.1.0-5	The HOST SYSTEM shall not corrupt/modify the Intrinsic Calibration Data. (this include ON time. if applicable the OFF time too) background: this data is used during video streaming. therefore shall not be corrupted	Requirement	IF	TBD
ECU_RS_359	**2.1.2**	**Camera Configuration Number**	**Heading**	**n/a**	**n/a**
ECU_RS_389	2.1.2.0-1	The "Camera Configuration Number" is set of data that needed to identify the correct drivers or configuration for Camera Head. Usually it is several bytes of data in a predefined structure that stored in a memory element.	Description	n/a	n/a
ECU_RS_423	2.1.2.0-2	For VT CH3 project: data stored in the EEPROM	Description	n/a	n/a
ECU_RS_501	2.1.2.0-3	WHEN: once in each life cycle before the video stream is used for safety related purposes (start-up. restarts)	Requirement	SM	TBD
ECU_RS_344	2.1.2.0-4	HOST SYSTEM shall read the "Camera Configuration Number" of the CAMERA HEAD and compare it with allowed values. If not match HOST SYSTEM shall remain in safe state.	Requirement	IF	TBD

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_352	2.1.2.0-5	The HOST SYSTEM shall read out and check the CRC Checksum of the "Camera Head Configuration Number". The address and data length in the EEPROM map is described in the following document: Document: [2] (Camera Interface Specification) Section: 5.3. EEPROM MAP Block Name: * Label Data Block \ "Bosch / Generic part" \ Camera Config Identifier * Label Data Block \ "Bosch / Generic part" \ Identification of Left/right/mono	Requirement	SM	TBD
ECU_RS_388	2.1.2.0-6	See HSI description how to read data from EEPROM memory of Camera Head.	Description	n/a	n/a
ECU_RS_360	**2.1.3**	**Unique Identification Number**	**Heading**	**n/a**	**n/a**
ECU_RS_441	2.1.3.0-1	The "Unique Identification Number" can be used to identify the Camera Head when required. Two typical use-cases: * The HOST SYSTEM receives video stream from multiple camera head and makes possible the identification of video source. * The Intrinsic Calibration Data stored permanent in the HOST SYSTEM instead of read out from Camera Head at each start-up. To ensure the connected Camera Head and the stored Calibration data match. the Unique Identification of Camera Head can be compared with the ID included in the Calibration Data.	Description	n/a	n/a
ECU_RS_502	2.1.3.0-2	WHEN: once in each life cycle before the video stream is used for safety related purposes (start-up. restarts)	Requirement	SM	TBD
ECU_RS_346	2.1.3.0-3	The HOST SYSTEM shall read the "Unique Indentification Number" and check for the allowed values. If not match HOST SYSTEM shall remain in safe state. The Unique Identification Number stored in the EEPROM. location described in the following document: Document: [2] (Camera Interface Specification) Section: 5.3. EEPROM MAP Block Name: ??	Requirement	IF	TBD
ECU_RS_354	2.1.3.0-4	The HOST SYSTEM shall check the CRC Checksum of the "Unique Identification Number" before uses its value. If not match HOST SYSTEM shall remain in safe state.	Requirement	SM	TBD
ECU_RS_363	**2.1.4**	**Configuration of Hardware Elements of Camera Head**	**Heading**	**n/a**	**n/a**
ECU_RS_438	2.1.4.0-1	The HOST SYSTEM shall perform the configuration of the following hardware components in the Camera Head: * Serializer: via I2C meessages that forwarded over the high-speed serial link * Imager: via I2C messages that forwarded over the high-speed serial link	Description	n/a	n/a
ECU_RS_367	**2.1.4.1**	**Imager Chip**	**Heading**	**n/a**	**n/a**
ECU_RS_331	2.1.4.1.0-1	The HOST SYSTEM shall permanently store (non volatile) the settings of Imager chip in the Camera Head. The config data shall not be corrupted in the HOST SYSTEM neither during OFF nor ON time.	Requirement	IF	ASIL B
ECU_RS_504	2.1.4.1.0-2	WHEN: always (ON and OFF time)	Requirement	IF	ASIL B
ECU_RS_384	**2.1.4.1.1**	**Check at Start-up or after config change**	**Heading**	**n/a**	**n/a**
ECU_RS_493	2.1.4.1.1.0-1	Info: there are groups of Imager registers: • STATIC Registers: used for configuring the Imager operation mode and these are not changed during video stream. • DYNAMIC register: used for fine-tune the image exposure (e.g.: analog gain. digital gain. exposure time etc...) Typically these are dynamically changed during during video streaming.	Description	n/a	n/a
ECU_RS_499	2.1.4.1.1.0-2	Validity of the section: STATIC register only.	Requirement	Other	ASIL B
ECU_RS_500	2.1.4.1.1.0-3	WHEN: start-up. restarts or operation mode change during runtime (e.g.: change from 60 fps to 30 fps)	Requirement	IF	ASIL B
ECU_RS_364	2.1.4.1.1.0-4	The HOST SYSTEM shall send the configuration data of Imager to Camera Head by I2C messages transferred via high speed serial link.	Requirement	IF	ASIL B

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_366	2.1.4.1.1.0-5	The HOST SYSTEM shall protect the sending of configuration data to Imager by insertion of CRC Checksum of I2C messages. note: it is safety mechanism to protect the Imager configuration during the data transmission from HOST SYSTEM to Imager chip.	Requirement	SM	ASIL B
ECU_RS_472	2.1.4.1.1.0-6	The HOST SYSTEM shall protect the sending of configuration data to Imager by insertion of "Message Counter" in addition to the CRC Checksum	Requirement	SM	ASIL B
ECU_RS_474	2.1.4.1.1.0-7	Details of the CRC Checksum and Message Counter: Document: [1] (Imager Safety Manual) Section: 6.1.11 [SM9. SM22] I2C CRC. I2C MESSAGE COUNTER Page: starts on page 60	Description	n/a	n/a
ECU_RS_473	2.1.4.1.1.0-8	This safety mechanism can be used in two way: 1) INTERNAL - Imager chip perform the comparison of "CRC Checksum" and "Message Counter" (called "Internal" in the Imager Safety Manual) 2) EXTERNAL - Imager shares the CRC Cheksum and the ECU performs the comparison with expected value (called "External" in the Imager Safety Manual) The VT CH3 SVC project uses as: TBD (internal/external)	Requirement	Other	ASIL B
ECU_RS_368	**2.1.4.2**	**Serializer Chip**	**Heading**	**n/a**	**n/a**
ECU_RS_332	2.1.4.2.0-1	HOST SYSTEM shall permanently store (non volatile) the settings of Serializer chip of Camera Head. The config data shall not be corrupted in the HOST SYSTEM neither during OFF time nor during ON time. WHEN: always (ON and OFF time)	Requirement	IF	ASIL B
ECU_RS_469	2.1.4.2.0-2	Info: the Serializer is used "Pixel Mode". it is a constrain from the deserializer (MAX96712) that does not support "Tunneling Mode".	Description	n/a	n/a
ECU_RS_386	**2.1.4.2.1**	**Check at Start-up or after config change**	**Heading**	**n/a**	**n/a**
ECU_RS_365	2.1.4.2.1.0-1	The HOST SYSTEM shall send the config data of Serializer to Camera Head via high speed serial link. (info: there is no microcontroller in the Camera Head that could perform the initialization of Serializer)	Requirement	IF	ASIL B
ECU_RS_505	2.1.4.2.1.0-2	WHEN: start-up. restarts or operation mode change during runtime (e.g.: change from 60 fps to 30 fps)	Requirement	IF	
ECU_RS_371	2.1.4.2.1.0-3	The HOST SYSTEM shall provide a safety mechanism to ensure the config register of Serializer are transmissed and written correctly. (for example: read back, transmission with CRC Checksum)	Requirement	SM	ASIL B
ECU_RS_448	**2.1.4.3**	**PMIC**	**Heading**	**n/a**	**n/a**
ECU_RS_461	2.1.4.3.0-1	The PMIC does not require configuration from HOST SYSTEM during the start-up of the Camera Head.	Description	n/a	n/a
ECU_RS_444	2.1.4.3.0-2	ERROR pin of the PMIC is connected to the Serializer but it is not used in the Functional Safety of the Camera Head. HOST SYSTEM can decide whether is it used for general error diagnostic.	Description	n/a	n/a
ECU_RS_459	**2.1.4.4**	**EEPROM**	**Heading**	**n/a**	**n/a**
ECU_RS_460	2.1.4.4.0-1	The EEPROM does not require configuration from HOST SYSTEM during the start-up of the Camera Head.	Description	n/a	n/a
ECU_RS_494	**2.1.5**	**Plausibility Check of Temperature Sensors**	**Heading**	**n/a**	**n/a**
ECU_RS_497	2.1.5.0-1	NOTE: currently this SM is under discussion. check the fesiblity from ECU side.	Description	n/a	n/a
ECU_RS_498	2.1.5.0-2	Goal: check the temperature values from the two different hardware element and ensure there failure of the temperature sensing.	Description	n/a	n/a
ECU_RS_506	2.1.5.0-3	WHEN: once in each life cycle before the video stream is used for safety related purposes (start-up. restarts)	Requirement	SM	TBD
ECU_RS_495	2.1.5.0-4	The HOST SYSTEM shall read the temperature value from the Imager and Serializer	Requirement	SM	TBD

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_496	2.1.5.0-5	HOST SYSTEM shall subtract the "Serializer temperature" from "Imager temperature" and compare ther result with the below threshold: Acceptable range for T_delta: -40 .. +40 [C]	Requirement	SM	TBD
ECU_RS_440	**2.16**	**Activites During Video Streaming**	**Heading**	**n/a**	**n/a**
ECU_RS_455	2.1.6.0-1	The mechanisms (either IF or SM) described in this section shall be activated before video streaming (more precisly when the video stream data is processing) and shall remain active during whole streaming phase.	Requirement	IF	ASIL B
ECU_RS_518	**2.1.6.1**	**Imager Rectify with Intrinsic Calibration Data**	**Heading**	**n/a**	**n/a**
ECU_RS_464	2.1.6.1.0-1	HOST SYSTEM shall use the Intrinsic Calibration Data to rectify the image frame. (continously for each image frames)	Requirement	IF	ASIL B
ECU_RS_361	**2.1.6.2**	**Image Synchronization**	**Heading**	**n/a**	**n/a**
ECU_RS_435	2.1.6.2.0-1	Validity of this section is depend on apllication whether uses the "Image Synchronization"? The function is based on the signal FRAME_SYNC generated by the HOST SYSTEM.	Description	n/a	n/a
ECU_RS_519	2.1.6.2.0-2	IF Image Synchronization is USED:	Description	n/a	n/a
ECU_RS_503	2.1.6.2.0-2.0-1	WHEN: continously during video stream. if it used for safety related purposes	Requirement	IF	
ECU_RS_351	2.1.6.2.0-2.0-2	The HOST SYSTEM shall correctly control the signal "FRAME_SYNC" that is used to synchronize the Image Capturing by the Camera Head. background: incorrect behaviour of this signal in the ECU will cause incorrect timing of the Image Capturing in the Camera Head (delayed picture) For specification (timing. pulse width. levels): see HSI	Requirement	IF	TBD
ECU_RS_436	2.1.6.2.0-3	IF Image Synchronization is NOT USED: the FRAME_SYNC signal shall be configure permantly to the state described in the Document [2]	Requirement	IF	TBD
ECU_RS_323	**2.1.6.3**	**CRC Protection for Image Data**	**Heading**	**n/a**	**n/a**
ECU_RS_324	2.1.6.3.0-1	The HOST SYSTEM shall check the CRC Checksum of the MIPI CSI2 Packets send by the Camera Head. CRC Checksum defined in the "Low Level Layer" of MIPI CSI2 Protocol. If the two CRC checksums are not equal. reject the image frame from further processing	Requirement	SM	ASIL B
ECU_RS_511	2.1.6.3.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_454	2.1.6.3.0-3	Further information in the following Section of Imager Safety Manual: Document: [1] (Imager Safety Manual) Section: 6.1.10 [SM8] Output Interface Fault Page: starts on page 59	Description	n/a	n/a
ECU_RS_347	2.1.6.3.0-4	If the Error Handling Algorithm detects an faulty operation the HOST SYSTEM shall ensure the safe state for the OVERALL SYSTEM	Requirement	SM	ASIL B
ECU_RS_350	**2.1.6.4**	**Frame Counter Monitoring**	**Heading**	**n/a**	**n/a**
ECU_RS_369	2.1.6.4.0-1	The HOST SYSTEM shall monitor the Frame Counter provided in the embedded data against the expected value. The value of counter is increased with +1 in each frame.	Requirement	SM	ASIL B
ECU_RS_512	2.1.6.4.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_431	2.1.6.4.0-3	Details of the Frame Counter provided by the Imager: Document: [1] (Imager Safety Manual) Section: 6.1.12 [SM10] Frame Counter Page: starts on page 65	Description	n/a	n/a
ECU_RS_432	2.1.6.4.0-4	If the Error Handling Algorithm of "Frame Counter" detects an faulty operation the HOST SYSTEM shall ensure the safe state for the OVERALL SYSTEM	Requirement	SM	ASIL B
ECU_RS_328	**2.1.6.5**	**Frame Rate Monitoring**	**Heading**	**n/a**	**n/a**
ECU_RS_452	2.1.6.5.0-1	This safety mechanism entirely base on the HOST SYSTEM. does not require information from the Camera Head.	Description	n/a	n/a
ECU_RS_508	2.1.6.5.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_329	2.1.6.5.0-3	The HOST SYSTEM shall monitor the the elapsed time between the received image frames.	Requirement	SM	ASIL B

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_330	2.1.6.5.0-4	If the elapsed time is shorter or longer than the defined value. signalling an ERROR: * High limit: YY ms (equivalent to frame rate: XYZ) * Low limit: YY ms (equivalent to frame rate: XYZ) Error Handling Algorithm: shall defined by Bosch or Volvo? (debouncing. threshold number of delayed frames etc...)	Requirement	n/a	ASIL B
ECU_RS_453	2.1.6.5.0-5	If the Error Handling Algorithm of "Frame Rate Monitoring" detects an faulty operation the HOST SYSTEM shall ensure the safe state for the OVERALL SYSTEM	Requirement	SM	ASIL B
ECU_RS_488	2.1.6.5.0-6	note: this Safety Mechanism shall be prepared for each frame rate option: 30 Hz and 60 Hz.	Description	SM	n/a
ECU_RS_439	**2.1.6.6**	**Imager - Exposure Control ([SM7-1] SFR monitor**	**Heading**	**n/a**	**n/a**
ECU_RS_382	2.1.6.6.0-1	The HOST SYSTEM shall check the exposure control data that sent in the embedded data of Imager for each frame that will be processed.	Requirement	SM	ASIL B
ECU_RS_509	2.1.6.6.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_451	2.1.6.6.0-3	Details of the data provide by Imager: Document: [1] (Imager Safety Manual) Section: 6.1.8 [SM7-1] SFR monitor Page: starts on page 53	Description	n/a	n/a
ECU_RS_383	2.1.6.6.0-4	The HOST SYSTEM shall check the CRC Checksum of the embedded data of Imager. To be clarify: is the CRC Checksum need to be checked.	Requirement	SM	ASIL B
ECU_RS_478	**2.1.6.7**	**Imager - [SM18-2] Test Pattern (ext)**	**Heading**	**n/a**	**n/a**
ECU_RS_479	2.1.6.7.0-1	The HOST SYSTEM shall check the fixed test pattern received in the embedded data of video stream transferred via MIPI interface.	Requirement	SM	ASIL B
ECU_RS_486	2.1.6.7.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_480	2.1.6.7.0-3	Description of the fixed pattern: Document: [1] (Imager Safety Manual) Section: 6.1.17 [SM18-2] TEST PATTERN (ext.) Page: 79	Description	n/a	n/a
ECU_RS_475	**2.1.6.8**	**Imager - [SM30] HFPN DETECTOR**	**Heading**	**n/a**	**n/a**
ECU_RS_476	2.1.6.8.0-1	The HOST SYSTEM shall implement the algorithm to detect HFPN (Horizontanl Fixed Pattern Noise) failure caused by internal random hardware fault of Imager chip.	Requirement	SM	ASIL B
ECU_RS_485	2.1.6.8.0-2	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_477	2.1.6.8.0-3	Description of the original safety mechanism: (this need to be replaced by the ECU) Document: [1] (Imager Safety Manual) Section: 6.1.26 [SM30] HFPN DETECTOR Page: 101	Description	n/a	n/a
ECU_RS_521	2.1.6.8.0-4	Details of HFPN Detector algorithm that shall be implemented in the HOST SYSTEM is in Document [3] An example code in Phyton is available.	Description	n/a	n/a
ECU_RS_463	**2.1.6.9**	**Serializer - Error Monitoring**	**Heading**	**n/a**	**n/a**
ECU_RS_507	2.1.6.9.0-1	WHEN: continously during video streaming when it is used safety related purposes	Requirement	SM	ASIL B
ECU_RS_372	2.1.6.9.0-2	The HOST SYSTEM shall check the ERRB pin or the dedicated error flags in register of the Deserilizer to detect failures signalled by the Serilizer. note: if the Serializer has an error it signalling for the Deserializer.	Requirement	SM	ASIL B
ECU_RS_487	2.1.6.9.0-3	If error is detected the HOST SYSTEM shall ensure the achivement of safe state.	Requirement	SM	ASIL B
ECU_RS_489	**2.1.7**	**Recovering from safe-states**	**Heading**		**n/a**
ECU_RS_447	2.1.7.0-1	To recover from CAMERA HEAD safe state the HOST SYSTEM shall switch OFF then ON the external power supply over the coaxial cable.	Requirement	IF	ASIL B
ECU_RS_362	**2.1.8**	**Other Requirements**	**Heading**	**n/a**	**n/a**

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_348	2.1.8.0-1	The HOST SYSTEM shall ensure the safe state of the OVERALL SYSTEM during the start-up of the Camera Head. (means before the Camera Head fully configured. the necessary start-up test are performed and starts the video stream)	Requirement	IF	ASIL B
ECU_RS_349	2.1.8.0-2	The HOST SYSTEM shall ensure the safe state of the OVERALL SYSTEM during shut-down of the Camera Head. Means when the nominal operation of Camera Head is not needed longer. the HOST SYSTEM shall change the OVERALL SYSTEM into safe state. and then can start to perform the shut-down sequence of the Camera Head.	Requirement	IF	ASIL B
ECU_RS_334	2.1.8.0-3	HOST SYSTEM shall not modify. overwrite or delete EEPROM Blocks other than those allocated for the HOST SYSTEM usage. WHEN: always (ON and OFF time)	Requirement	IF	ASIL B
ECU_RS_335	**2.1.9**	**Environment Conditions**	**Heading**	**n/a**	**n/a**
ECU_RS_513	2.1.9.0-1	WHEN: always (ON time and OFF Time)	Requirement	IF	ASIL B
ECU_RS_336	2.1.9.0-2	The HOST SYSTEM shall guarantee the environemental and electrical conditions specified in the Volvo TR. For examples (but not a complete list): * ambient temperature * sun exposure * vibration * exposure to water or any other liquid penetration * exposure to chemicals or other substances * life time and operation profile	Requirement	IF	ASIL B
ECU_RS_339	2.1.9.0-3	Power supply for Camera Head (Power Over Coax) is according to the specs.	Requirement	IF	ASIL B
ECU_RS_465	2.1.9.0-4	The power supply for CAMERA HEAD provided by HOST SYSTEM over coaxial cable shall fulfill the requirements defined in the Document [2] - Camera Interface Specification	Requirement	IF	ASIL B
ECU_RS_449	2.1.9.0-5	Power supply of Camera Head shall not be directly connected to the vehicle 12V battery voltage. Rational: The camera head is not designed to be supplied by vehicle battery voltage and has not not protection measures in the Camera Head. E.g. not robust against battery dynamic behaviour. pulses. reverse polarity etc.	Requirement	IF	ASIL B
ECU_RS_285	**2.3**	**Requirements for the Product Life Cycle**	**Heading**	**n/a**	**n/a**
ECU_RS_278	**2.3.1**	**Production**	**Heading**	**n/a**	**n/a**
ECU_RS_280	2.3.1.0-1	List here the project specific production requirements which can be - requirements from the production plant at the OEM / Bosch plant. or - requirements for manufacturability - requirements which enable verification measures during production - logistic requirements (e.g. packaging. delivery. etc.)	Description	n/a	n/a
ECU_RS_370	2.3.1.0-2	The TCD (Technical Customer Documentation) shall be followed.	Requirement	IF	
ECU_RS_244	**2.3.2**	**Service**	**Heading**	**n/a**	**n/a**
ECU_RS_355	2.3.2.0-1	The Camera Head does not require regular or occasional maintenance.	Description	n/a	n/a
ECU_RS_356	2.3.2.0-2	Any kind of disassembly or repair of the Camera Head is NOT permitted. In case of failure the complete Camera Head shall be replaced.	Requirement	IF	ASIL B
ECU_RS_276	**2.3.3**	**Decomissioning**	**Heading**	**n/a**	**n/a**
ECU_RS_357	2.3.3.0-1	The power supply shall be switched OFF before connects or disconnects the coaxial cable of the Camera Head.	Requirement	IF	ASIL B
ECU_RS_427	**3**	**Referenced Documents**	**Heading**	**n/a**	**n/a**
ECU_RS_428	3.0-1	Reference ID: [1] Description: Safety Manual of Imager Title: Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM Format: PDF (eletronics) Pages: 141 (pages in the PDF) Revision: v1.0.1.0 Revision Date: May 30. 2025	Description	n/a	n/a

ID	Object Number	ECU FuSa for the Customer	Object_Type	DA_Fusa_ReqType	Safety_Integrity
ECU_RS_433	3.0-2	Reference ID: [2] Description: Camera Interface Specification Author: Bosch Title: Camera Interface Specification Format: PDF (eletronics) Pages: Revision: Revision Date:	Description	n/a	n/a
ECU_RS_522	3.0-3	Reference ID: [3] Description: Algorithm for HFPN Detection Author: Samsung Title: Detection of HFPN Potentials on S5K3B6S Format: PDF (eletronics) Pages: 24 Revision: v1.0.1.0 Revision Date: May 30. 2025	Description	n/a	n/a
ECU_RS_456	**4**	**Document History**	**Heading**	**n/a**	**n/a**
ECU_RS_520	4.0-1	DOORS Baseline: 17.2 Date: 2025-10-22 (yyyy-mm-dd) Author: Tibor Rácz (XC-AC/ENH2-Bp) Status (draft/reviewed/released): DRAFT Description/Changes: Change wording. re-structure for more logical overview.	Description	n/a	n/a